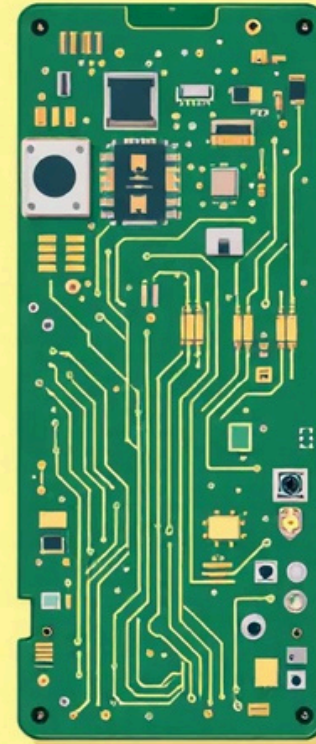


AI-Powered PCB Defect Detection using EfficientNet and Image Processing

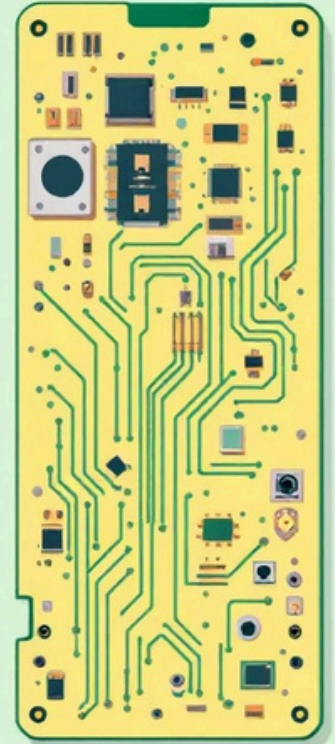
Automated inspection of PCB defects through deeplearning and computer vision.

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Perfect
PCB



Defected
PCB



The Problem: Manual PCB Inspection

Manual inspection of Printed Circuit Boards(PCBs) is inherently **time-consuming and prone to human error**. This traditional approach directly impacts production efficiency and product quality.

1

Challenges in Detection

Distinguishing minute visual differences between "golden" (perfect) and defective images proves challenging, especially with multiple defect types.

2

Goal: Automated Defect Detection

Our objective is to implement an automated system that ensures consistent quality, significantly increases inspection speed, and reduces operational costs.

Class Labels:

Missing Hole

Mouse Bite

Open Circuit

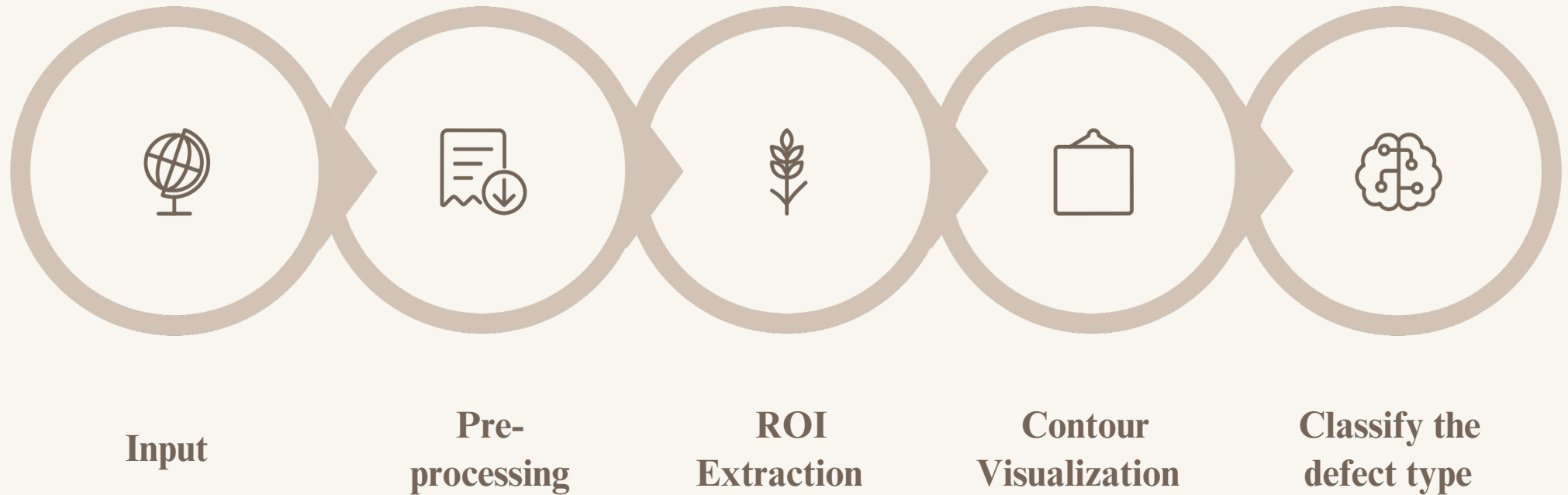
Short

Spur

Spurious Copper

System Architecture: From Input to Insight

AI-powered system processes PCB images through a structured pipeline, transforming raw data into actionable defect classifications



Each stage is meticulously designed to enhance detection accuracy and provide a comprehensive analysis of PCB integrity

Methodology: Core Techniques Employed

A blend of advanced image processing and deep learning techniques underpins the system's ability to accurately identify and classify PCB defects.



Image Subtraction

Precisely highlights defective areas by comparing golden and defected images pixel by pixel.



Contour Detection

Utilises `cv2.findContours()` to precisely localise Regions of Interest (ROIs) around detected anomalies.



Deep Learning

EfficientNet-B4, enhanced with a custom classifier head, performs robust defect type prediction.



Streamlit Frontend

Provides an intuitive, interactive dashboard for seamless defect detection and result visualisation.

Interactive Defect Detection with Streamlit

A user-friendly Streamlit application provides an interactive interface for uploading PCB images and visualising detection results in real-time.

01

Upload Images

Users can easily upload both golden and potentially defected PCB images

03

View Results

Displays the subtraction image, ROI bounding boxes, and the classified defect type with confidence score.

02

Initiate Analysis

Clicking "Done" triggers the full defect detection pipeline.

04

Download Processed Images

Option to download the annotated images for further analysis or record-keeping.

Model Training and Evaluation

The EfficientNet-B4 model was rigorously trained and validated to achieve high accuracy in defect classification.

Implementation Details:

- Framework: PyTorch + timm for model management.
- Model Architecture: EfficientNet-B4, renowned for efficiency and performance.
- Input Size: Images resized to 300x300 pixels for consistent processing.
- Optimiser: Adam, selected for its adaptive learning rate capabilities.
- Training Strategy: Early Stopping and Checkpoints to prevent overfitting and save best models.

Results Summary:

The model demonstrated strong performance across validation and test datasets.

Validation Accuracy	92%
Test Accuracy	87%

These metrics confirm the model's robustness in identifying PCB defects.

Key Observations and Conclusion

The system demonstrates strong capabilities in identifying common PCB defects, providing precise localisation and classification.

Accurate Defect Detection	HighConfidencePredictions	Challenging Defect Types
The model accurately detects major defects such as Missing Holes, Short Circuits, and Open Circuits, which are critical for PCB functionality.	For well-defined PCB data, the system consistently provides high-confidence predictions, enhancing reliability.	Some misclassifications were observed for "Spur" and "Spurious Copper," primarily due to their subtle visual similarities and potential ambiguities.

This solution provides a robust and efficient alternative to manual inspection, significantly improving quality control in PCB manufacturing